

KHOA NGUYEN

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RESEARCH INTEREST

My research focuses on building **reliable** and **privacy-preserving agentic systems** for real-world security and software environments. My work lies at the intersection of **LLM systems, tool-augmented agents, distributed learning, and differential privacy**, aim to enabling AI systems that can reason over complex data, interact with external tools, and operate safely in data-sensitive settings. My work centers on two directions:

- **Privacy-preserving LLM and agent systems:** bridging formal theoretical guarantees with empirical verification to design scalable defense mechanisms, utilizing local differential privacy, that protect prompts, latent states, and multi-step outputs against carefully designed reconstruction, extraction, and inference attacks under realistic adversarial settings.
- **Reliable agent deployment at industry scale:** translating foundational research into production-grade systems by leveraging distributed tuning and resource optimization to improve the scalability, controllability, and verified robustness of agentic workflows across security-critical software environments.

EDUCATION

New Jersey Institute of Technology *Ph.D in Data Science* *Sep 2023 – Present*
Focus: LLMs, VLMs, Post-Training (SFT & RL), Tool-augmented, Agentic Systems, Security, Privacy, Fuzzing.
New Jersey Institute of Technology *Sep 2019 – May 2023*
B.S. in Computer Science

INDUSTRY EXPERIENCE

OppyAI Inc. *Nov 2024 – Present*
Research Scientist

- Translated one of my core research projects into a commercial startup, designing privacy-preserving mechanisms for LLMs that secure sensitive data during scalable training and inference.
- Developed end-to-end distributed ML pipelines for large-scale model development, spanning data preprocessing, pre-training, SFT/RL post-training, inference, and evaluation on multi-node, multi-GPU infrastructure.

AT&T *Nov 2025 – Present*
Research Collaborator

- **Cellular Network Maintenance Scheduling:** Designed a graph foundation model to predict cellular sites maintenance, optimizing for both operational efficiency and user experience. Processed and analyzed large-scale historical LTE and 5G network data to train and evaluate scheduling systems.

TECHNICAL SKILLS

Machine Learning: LLMs, VLMs, NLP, RL, Distributed Training, GNN, Federated Learning, Distillation.

Frameworks: Pytorch, Transformers, Llama-Factory, verl, vllm, Ray, Triton, DeepSpeed, Accelerate.

Infrastructures & Tools: CUDA, Linux, Slurm, Git, Docker, SQL.

Programming Languages: Python, Java, JavaScript, C/C++.

RESEARCH PROJECTS

Privacy-Preserving Generation with LLMs *Jan 2024 – Present*

- **Project Goals:** Develop privacy-preserving agentic systems that safeguard sensitive user information while preserving strong model utility, with a focus on reliable deployment in interactive, tool-augmented, and agent-based settings.
- **Methodology:** Developed NOIR, a split-learning LLM framework with local differential privacy (LDP); analyzed reconstruction risk through theoretical bounds on attack recovery; and extending the mechanism toward multimodal, tool-use, and agentic systems.

Program Structure-Aware LLM for Targeted Software Testing

Jan 2025 – Present

- **Project Goals:** Enhance the reliability and controllability of agentic systems for software testing through targeted test-case generation aligned with specific program behaviors, supporting improved bug discovery, software assurance, and security analysis.
- **Methodology:** Designed a program structure-aware framework that conditions agentic systems on program representations to guide behavior-specific test generation, achieving higher precision and effectiveness compared to strong baseline methods.

FedChip: Federated LLM for AI Accelerator Chip Design

Apr 2024 – Apr 2025

- **Project Goals:** Enable collaborative LLM-based hardware design across distributed participants without centralizing sensitive design data, improving both practicality and design quality in multi-party settings.
- **Methodology:** Developed a federated framework for AI accelerator chip design, constructed a multi-objective evaluation pipeline for design quality, and demonstrated substantial improvements over prior LLM-based hardware design methods.

Geographic-Aware Federated Learning

Sep 2023 – Apr 2024

- **Project Goals:** Improve the scalability and effectiveness of federated learning in decentralized environments by modeling structured relationships among participants under realistic data heterogeneity.
- **Methodology:** Developed a geographic-aware federated learning framework that incorporates real-world relational structure across clients, and validated its effectiveness on large-scale distributed datasets.

PUBLICATIONS

1. K. Tran, **Khoa Nguyen**, C. Borcea, N. Phan. *Program Structure-aware Language Models: Targeted Software Testing beyond Textual Semantics*. Findings of the Association for Computational Linguistics (**ACL'26**).
2. **Khoa Nguyen***, K. Ton*, N. Phan, and et al. *NOIR: Privacy-Preserving Generation of Code with Open-Source LLMs*. The USENIX Security Symposium (**USENIX Security'26**).
3. **Khoa Nguyen***, K. Tran*, N. Phan, and et al. *SGFusion: Stochastic Geographic Gradient Fusion in Federated Learning*. International Conference on Big Data (**BigData'25**).
4. M. Nazzal*, **Khoa Nguyen***, D. Vungarala*, and et al. *FedChip: Federated LLM for Artificial Intelligence Accelerator Chip Design*. IEEE International Conference on LLM-Aided Design (**ICLAD'25**).

PATENTS

1. N. Phan, **Khoa Nguyen**, K. Ton, and et al. *System and Method for Private Generation of Code*. International Patent Application, No. PCT/US25/042391.

SERVICE

- Teaching Assistant at NJIT: DS110: Basic Foundations of AI, CS280: Programming Language Concepts, DS677: Deep Learning, DS636: Data Analytics with R Program.

ACHIVEMENTS

Winner – GfK's NextGen Data Science Hackathon	2022
ICPC – Qualified for Greater NY Regional Contest	2022
3rd Prize – Vietnam Science and Engineering Fair (ViSEF)	2017